

Notes: Rossi and Godard (AEJ: Policy, 2022)
**The Old-Age Security Motive for Fertility: Evidence from the Extension of
Social Pensions in Namibia**

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1 Big picture

- **Question.** Do old-age pensions reduce fertility because they weaken the need to rely on children for support in old age?
- **Setting.** The paper studies Namibia in the 1990s, when the post-apartheid government equalized social pension benefits across ethnic groups and expanded coverage across regions. This reform removed both ethnic differences in generosity and regional differences in access.
- **Core challenge.** A fall in fertility after pension expansion could reflect many other changes happening around independence: urbanization, poverty reduction, local development, AIDS, migration, or a broader convergence in fertility across groups. The paper must isolate the pension channel from these concurrent structural changes.
- **Main contribution.** The paper combines pre-reform variation in pension generosity and coverage with cohort exposure to the reform. It studies both birth timing and completed fertility, and it explicitly asks whether pensions matter most in late reproductive life, where the old-age-security motive should be strongest.
- **Main baseline result.** Social pensions substantially reduce fertility. In the period specification, the difference-in-differences coefficient becomes about -0.17 once controls are added. Given the observed gap in initial needs, this implies a differential decline of roughly 0.65 children between high- and low-needs groups.
- **Main age-pattern result.** The effect is concentrated among women older than 25, especially those between 30 and 45. Women respond mainly by stopping earlier, not by delaying the first birth.
- **Main cohort result.** Younger cohorts exposed to the equalized pension system have fewer children than older cohorts, and the decline is stronger where pre-reform needs for informal old-age support were higher. The cohort estimates imply a differential decline of about 0.5 to 0.75 children between high- and low-needs groups.
- **Main mechanism result.** The paper presents evidence that pensions change desired fertility. Ideal family size falls, the fraction of women who want to stop having children rises, and age at last birth converges downward in the initially low-coverage regions.
- **Main nonlinearity result.** The fertility effect is strongest where initial protection was very low. Moving from roughly one-third coverage to one-half coverage lowers fertility by about one child, while further increases in protection have little detectable effect.
- **Economic takeaway.** The paper gives rare quasi-experimental evidence for the old-age-security motive for fertility and suggests that expanding social pensions could materially accelerate fertility decline in sub-Saharan Africa.

2 Data and measurement (key objects)

2.1 Why Namibia is a useful laboratory

- Namibia inherited a highly unequal pension system from apartheid. Before reform, pension benefits differed by ethnicity and effective coverage differed sharply by region.
- The 1990s reform equalized the system in stages: benefits were equalized in law in 1992, and coverage became close to universal after outsourcing delivery in 1996.
- The reform therefore created within-country variation that looks like a cleaner natural experiment than cross-country or simple before-after comparisons.
- The setting is especially attractive because direct data exist on birth histories, allowing the paper to study both when women give birth and how many children they ultimately have.

Interpretation. The paper is not just exploiting that pensions exist. It exploits that the old system disadvantaged some ethnic-region groups much more than others, and that the new system removed those inequalities quickly.

2.2 Data sources

The paper combines three data sources.

- **DHS fertility data.** Demographic and Health Surveys for 1992, 2000, 2007, and 2013 provide complete birth histories for women. These data allow the authors to reconstruct annual birth rates and cohort completed fertility.
- **NHIES socioeconomic data.** Namibia’s Household Income and Expenditure Survey for 1993–1994, 2003–2004, 2009–2010, and 2015–2016 provides regional and ethnic information on poverty, urbanization, financial inclusion, schooling, child labor, and access to health services.
- **Regional HIV data.** The authors add regional HIV prevalence from the 1994 National HIV prevalence survey.

The core treatment cell is the **region** $\times$ **ethnicity cluster**. After dropping Whites and the Khomas region, the final analysis uses 35 clusters in the period regressions and 26 or 28 clusters in the cohort regressions depending on the sample.

2.3 How the treatment is measured

The paper builds the treatment in two steps.

Step 1: expected pension. Before reform, expected pension in a cluster is defined as

$$ExpectedPension_c = Coverage_c \times OfficialBenefit_c. \quad (1)$$

Coverage varies mainly by region, while official benefits vary mainly by ethnicity.

Step 2: initial needs for old-age support. The main explanatory variable is

$$InitialNeeds_c = PovertyLine - ExpectedPension_c. \quad (2)$$

The poverty line is 660 rands per year. Larger *InitialNeeds* means weaker formal old-age protection and therefore greater need to rely on children.

Interpretation. The treatment is not “pension receipt” at the individual level. It is the ex ante need for informal support in old age implied by the cluster’s pre-reform pension environment.

2.4 Pre-reform variation in benefits and coverage

- **Benefits by ethnicity.** Before the extension, Owambo, Kavango, and Caprivi were entitled to only 55 rands per month; Herero and Nama to 65; Damara to 75; and Whites to 382. After the reform, everyone received 160.
- **Coverage by region.** Effective coverage among Blacks ranged from roughly 31% to 65% across the 12 relevant non-capital regions before reform. By 1998 national coverage had jumped to 88%, and by 2001 it was about 95%.
- **Expected pension range.** In the final analysis sample, expected annual pension varies from about 200 to 585 rands across clusters.

Interpretation. This two-dimensional variation is the heart of the design. Some groups were disadvantaged because the law gave them low benefits; others because the administration failed to deliver pensions reliably.

2.5 Outcome variables

The paper studies fertility in two complementary ways.

Period analysis (tempo).

$$Birth_{i,t} = 1 \quad \text{if woman } i \text{ gave birth in year } t. \quad (3)$$

The period outcome is the annual probability of birth.

Cohort analysis (quantum).

$$BirthNumber_i = \text{total number of children ever born to woman } i. \quad (4)$$

The cohort outcome is completed fertility for women aged 44 to 50 at survey date.

The paper also studies mechanism outcomes:

- reported ideal number of children,
- whether a woman wants to stop childbearing,
- age at last birth,
- age at first birth.

Interpretation. The period specification tells us *when* fertility changes. The cohort specification tells us whether those timing shifts translate into fewer total children.

2.6 Controls

The full set of controls includes:

- **urban structure:** share living in urban areas and share active in agriculture,
- **wealth and finance:** poverty rate and share with bank or savings account,
- **access to services:** share living more than one hour from a hospital or clinic,
- **cost of children:** school attendance of ages 5–14 and child labor of ages 10–14,
- **health environment:** HIV prevalence and under-5 mortality,
- **individual controls:** maternal education.

These controls are measured before the extension and interacted with the post or exposure variable.

Interpretation. The controls are designed to absorb exactly the structural factors that might otherwise explain a differential fertility decline unrelated to pensions.

2.7 Sample restrictions

A crucial empirical choice is to exclude:

- **Whites**, who had very different pre-reform treatment, trends, and socioeconomic status,
- **Khomas**, the capital region around Windhoek, where Blacks had exceptionally low pension access but very different development dynamics.

Interpretation. The preferred sample is intentionally narrower than the whole country. The goal is credibility, not representativeness of every Namibian subgroup.

2.8 Key descriptive facts

- Before reform, fertility was much higher in low-coverage regions than in high-coverage regions: roughly six versus five children per woman in 1992.
- The pre-reform gap was concentrated at older ages. Women in low-coverage regions did not start earlier; they stopped later.
- After the extension, fertility fell everywhere, but especially in the initially disadvantaged regions. By the mid-2000s the gap had essentially disappeared.
- Among Blacks outside Khomas, high- and low-coverage regions are much closer in observables than they appear in the full sample including Whites.

Interpretation. The raw data already point toward the mechanism: the reform mattered most where the old-age-security motive should matter most, namely for women in late reproductive life who had not yet stopped childbearing.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper’s central hypothesis is that parents have more children when they need informal support in old age. A pension reduces that need and therefore lowers desired fertility. This logic yields three empirical predictions:

- groups with larger pre-reform needs should reduce fertility more after the pension expansion;
- the effect should be strongest late in reproductive life, because stopping behavior responds faster than the age at first birth;
- if the mechanism is truly about old-age security, the evidence should look more like a change in desired completed fertility than like improved contraceptive control alone.

3.2 Baseline period specification

The main mother-year panel regression is

$$Birth_{i,t} = \gamma InitialNeeds_{c(i)} \times Post_t + \beta_t + \alpha_{c(i)} + \nu Z_i + \delta X_{c(i),pre} \times Post_t + \mu_{w(i)} + \varepsilon_{i,t}. \quad (5)$$

Here:

- $Birth_{i,t}$ is a birth dummy,
- $Post_t$ equals one after 1992,
- β_t , α_c , and μ_w are year, cluster, and survey-wave fixed effects,
- Z_i are time-invariant individual controls,
- $X_{c,pre}$ are pre-reform cluster controls interacted with $Post_t$.

Interpretation. The coefficient γ asks whether fertility fell more after 1992 in groups that had higher unmet old-age-support needs before reform.

3.3 Baseline cohort specification

The cohort-level regression is

$$BirthNumber_i = \gamma InitialNeeds_{c(i)} \times Exposure_{k(i)} + \beta_{k(i)} + \alpha_{c(i)} + \nu Z_i + \delta X_{c(i),pre} \times Exposure_{k(i)} + \varepsilon_i. \quad (6)$$

The exposure variable is constructed as

$$Exposure_k = \max(\min(10, k - 1955), 0) / 10, \quad (7)$$

so cohorts born before 1955 are unexposed, cohorts born after 1965 are fully exposed, and cohorts born in between are partially exposed.

Interpretation. This design uses the fact that women were at different ages when the new pension system arrived. If stopping behavior is the margin that matters, then cohorts in their 30s during the reform should respond most.

3.4 Estimation details

- The period specification uses retrospective birth histories to construct annual births from 1990 to 2012 for women aged 15 to 40 in year t .
- The 1992 DHS wave is not used in the period regression because the region coding is too coarse there.
- The cohort specification restricts to women aged 44 to 50 to reduce censoring of completed fertility.
- Regressions use survey weights.
- Standard errors are clustered at the region $\times$ ethnicity level, which is the treatment level.

Interpretation. The paper squeezes a lot of identifying variation out of retrospective survey histories while remaining careful about age composition and treatment clustering.

3.5 Age heterogeneity in the period analysis

The authors estimate the period DD separately by age group. The key finding is that the interaction coefficient is negative only for women older than 25, and especially relevant for ages 30 to 39.

Interpretation. This is exactly what the old-age-security story predicts. A pension should not mainly operate by delaying entry into motherhood. It should operate by changing how late women continue to have children.

3.6 Nonlinear specification

To test whether the effect is linear, the paper replaces continuous *InitialNeeds* with three categories:

- low needs,
- medium needs,
- high needs,

where medium is the omitted category.

Interpretation. If the relationship were linear, the low- and high-needs coefficients should be symmetric around zero. Instead, the high-needs coefficient is large and significant while the low-needs coefficient is close to zero, which implies a strongly nonlinear effect of pension protection on fertility.

3.7 Mechanism regressions

The mechanism analysis keeps the same difference-in-differences logic but changes the outcome variable to fertility preferences or stopping-timing margins. The authors look for convergence in:

- ideal family size,
- intention to stop childbearing,

- age at last birth,
- age at first birth.

Interpretation. A pure birth-control story would most naturally affect unwanted births or early timing. The paper instead finds evidence of a shift in desired completed family size and stopping behavior.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

The paper does not use an external instrument in the usual IV sense. Identification comes from:

- differences in *pre-reform* pension generosity and coverage across clusters,
- timing of the reform,
- differences in *cohort exposure* to that reform,
- rich fixed effects and pre-reform controls.

Interpretation. The baseline question is: once the pension system is equalized, do the groups that were initially most underprotected reduce fertility more than the groups that were already relatively protected?

4.2 Why reverse causality is weak

A common concern is that falling fertility could itself generate pension reform. The authors argue this is less plausible here because the extension was driven by post-apartheid equity motives: the explicit objective was to grant the same rights to everyone, not to respond to local fertility dynamics.

Interpretation. This helps with one of the hardest issues in the broader pensions-and-fertility literature. The reform was politically about equal rights, not about targeting areas where fertility had already started to decline.

4.3 Parallel trends and the exclusion of Whites and Khomas

The paper's most important identifying assumption is that, absent the reform, fertility would have evolved similarly across the relevant groups. Table 4 shows why the sample restriction matters.

- Whites have a distinct pre-trend: fertility was already declining before the extension.
- Khomas also has distinct pre-trends and post-trends because the capital region followed a different development path.
- Once Whites and Khomas are excluded, pre- and post-trends are essentially uncorrelated with *InitialNeeds*, while the differential decline appears only during the extension period 1992–1997.

Interpretation. The sample restriction is not cosmetic. It is what turns a suggestive comparison into a more credible DD design.

4.4 Why omitted variables likely bias the estimates downward

Adding controls for urbanization and wealth makes the coefficient more negative. The paper's explanation is intuitive:

- regions with initially high pension coverage were urban but poor in the apartheid era,
- after independence, these regions benefited more from structural transformation,
- that faster urbanization and poverty reduction should itself reduce fertility,
- so failing to control for these trends masks part of the pension effect.

Interpretation. This is a rare case where omitted variables appear to create attenuation bias rather than exaggerating the headline result.

4.5 Homogeneous treatment effect in a fuzzy DD

Because the treatment is group-level need for informal old-age support rather than an individual binary treatment, the authors discuss the homogeneity requirement emphasized in fuzzy DD work. They argue this assumption is plausible only in the final analysis sample, where high- and low-coverage Black regions outside Khomas are similar on observables and converge to similar fertility levels after reform.

Interpretation. The paper is explicit that this is not a perfect design. Its identification rests on choosing the subset of groups for which the treatment effect is plausibly comparable.

4.6 Migration and composition effects

The paper spends substantial effort ruling out that the catch-up is due to people moving rather than fertility changing.

- Migrants and natives have similar fertility in both low- and high-coverage regions.
- Excluding recent migrants leaves the coefficients broadly unchanged.
- Because the main regressions use retrospective birth histories, out-of-sample migration cannot explain the pre-1992 relationship between needs and fertility among women observed later.
- A mother fixed-effects version of the period analysis remains negative, although less precise because identification becomes weaker.

Interpretation. The migration checks matter because Namibia had substantial mobility after apartheid. The paper shows that mobility does not line up with the observed fertility convergence in the way needed to explain the results away.

4.7 What the paper can and cannot distinguish

The paper argues that the old-age-insurance channel is the most convincing interpretation, but it is careful not to overclaim.

- **Against a female-labor-supply story:** there is no catch-up in female labor-force participation between high- and low-coverage regions.
- **Against a pure income story:** cash transfers alone do not naturally explain lower completed fertility among older women, and the negative effect is stronger in households with no elderly pension recipient present.
- **Still, not fully pinned down:** pensions may also affect household resources and intergenerational sharing, so the insurance mechanism is favored but not observed directly.

Interpretation. The paper is strongest on the reduced form — pensions lower fertility — and on the stopping-margin mechanism. It is somewhat less definitive on the exact decomposition between insurance and other household channels.

5 Tables and figures

The paper’s empirical argument moves: first it documents the unequal pension system; then it shows raw fertility convergence; next it validates the difference-in-differences design; after that it establishes the main period and cohort results; and finally it shows that the mechanism is stopping rather than delayed starts.

5.1 Figure 1 and Tables 1–2: why Namibia creates usable identifying variation

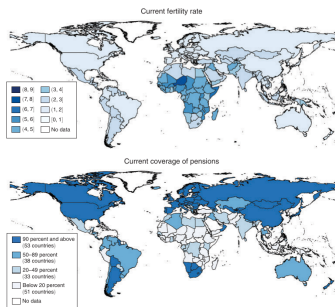


FIGURE 1. FERTILITY AND PENSIONS IN THE WORLD
Notes: The first map shows the average number of births per woman in 2017, based on estimates from United Nations (2017). The second map shows the percentage of the population above statutory pensionable age receiving a pension.

TABLE 1—VARIATION IN PENSION BENEFITS ACROSS ETHNIC GROUPS

	Pension per month (1)	Ratio to poverty line (2)	Ratio to average income (3)	Fraction of population (4)
<i>Before the extension</i>				
White	R382	7.0	0.96	6%
Coloured	R192	3.5	0.48	6.5%
Baster	R150	2.8	0.38	1%
Tswana	R109	1.8	0.25	0.5%
Damara	R75	1.4	0.20	10%
Herero, Nama	R65	1.2	0.16	12%
Ovambo, Kavango, Caprivi	R55	1	0.14	64%
<i>After the extension</i>				
All	R160	3	0.40	100%

Notes: The table shows the official amount of social pensions paid to different ethnic groups, in absolute terms and relative to the poverty line and the average income, before (1989) and after (1996) the Pension Act.

Source: Deveness (2001).

TABLE 2—VARIATION IN PENSION COVERAGE ACROSS REGIONS

		Fraction of recipients among eligible people		Fraction of population
		All (1)	Blacks only (2)	(3)
<i>Before the extension</i>				
South region	Hardap	78	48	4.7
	Karas	76	61	4.4
	Erongo	72	57	3.8
Central region	Khomas*	66	15	11.9
	Caprivi	65	65	6.3
Northeast region	Omahenge	60	52	3.7
	Opowandjupa	48	36	7.2
North region	Kaunene	51	45	4.6
	Oshikoto	49	46	9.1
	Karas	39	38	8.4
	Ohangwena	36	35	12.8
	Onamati	34	34	13.6
Namibia	Oshana	31	31	9.5
		49	42	100
<i>After the extension</i>				
Namibia		88	n/a	100

Notes: The table shows the effective coverage of social pensions in different regions before (1993–1994) and after (1995) the reintroduction of pension delivery, for all ethnic groups and for Blacks only. *Khomas is the region of the capital city (Windhoek).

Source: all ethnic groups (Subbarao 1999); Blacks in 1993–1994, own computation based on the

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- Figure 1 shows a broad negative association worldwide between fertility and pension coverage, motivating the question.
- Table 1 shows just how unequal the pre-reform Namibian pension system was: monthly benefits ranged from 55 rands for the least favored Black groups to 382 for Whites.

- Table 2 shows strong regional inequality in effective coverage before outsourcing, with Black coverage roughly between 31% and 65% across the main non-capital regions.

Economic message. The paper is built on the fact that Namibia moved from a highly unequal system to an almost universal one. Without that pre-reform variation in both benefits and delivery, there would be no clean within-country design.

5.2 Table 3 and Figure 2: the raw fertility gap is real, and it closes after the reform

TABLE 3—DESCRIPTIVE STATISTICS BEFORE THE EXTENSION OF PENSIONS

	All			Blacks only		
	High coverage (1)	Low coverage (2)	Khomas (3)	High coverage (4)	Low coverage (5)	Khomas (6)
<i>Panel A. Individual characteristics</i>						
Ethnicity						
% European/Afrikaans	0.15	0.00	0.39	—	—	—
% Damara	0.25	0.01	0.22	0.29	0.01	0.36
% Herero	0.18	0.00	0.09	0.21	0.00	0.15
% Owambo/Kavango/Caprivi	0.42	0.98	0.30	0.49	0.99	0.49
Education (higher than primary)	0.43	0.31	0.35	0.33	0.31	0.28
<i>Panel B. Regional controls</i>						
% poor	0.32	0.38	0.07	0.36	0.38	0.10
% living in urban areas	0.33	0.10	0.88	0.27	0.09	0.83
% econ. active in agriculture	0.44	0.59	0.05	0.50	0.60	0.07
% with bank/saving account	0.40	0.28	0.68	0.36	0.28	0.64
% 10–14 economically active	0.07	0.07	0.00	0.08	0.07	0.00
% 5–14 ever in school	0.86	0.93	0.94	0.84	0.93	0.93
% with hospital/clinic > 1 hour walk	0.51	0.50	0.15	0.56	0.50	0.14
Prevalence of HIV	0.10	0.09	0.07	0.10	0.09	0.07
Child mortality rate	0.08	0.06	0.05	0.08	0.06	0.05
Observations NHIES	11,204	10,874	2,906	9,194	10,841	1743
Observations DHS	2,711	2,149	561	2,213	2,137	499

Notes: The table displays the mean of important control variables, separately for regions with low (North) initial pension coverage, regions with high (Northeast, South, and Erongo) initial pension coverage, and the region of the capital city Khomas. In the first three columns, all ethnic groups are included. In the last three columns, only Blacks are included. The sample of interest for our analysis excludes Whites and Khomas, and corresponds to columns 4 and 5. Table A.2 in the online Appendix indicates the statistical significance of pairwise differences. The variables are constructed as follows: education is a dummy equal to 1 if the mother completed more than primary school in DHS 1992; urban areas are defined by the Local Authorities Act No23 1992; agricultural sector is defined according to the classification of activities in NHIES 1993–1994; a household is classified as poor if they reported spending more than 60 percent of their budget on food in NHIES 1993–1994; the schooling rate measures the share of

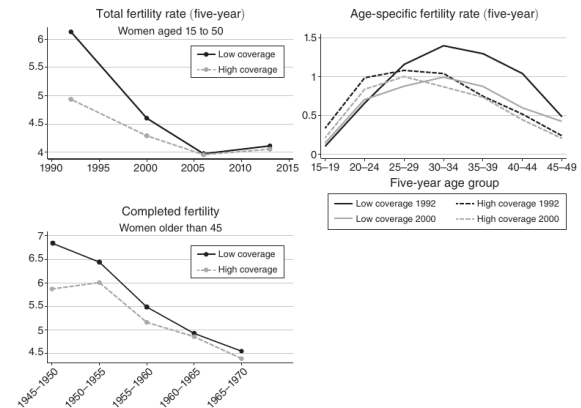


FIGURE 2. EVOLUTION OF FERTILITY, BY INITIAL PENSION COVERAGE

Notes: The graphs plot the evolution of several measures of fertility in regions with low (North, solid lines) and high (Northeast and South, dashed lines) initial pension coverage. The extension of pensions took place between 1992 and 1997. Data: DHS. Sample: non-White women aged 15 to 50 in the fertility rates graphs, older than 45 in the completed fertility graph. The graph on the top left shows Total Fertility Rate, a synthetic rate measuring how

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- Table 3 shows that the full sample is badly imbalanced because Whites and Khomas are much richer and more urban. Once the sample is restricted to Blacks outside Khomas, the high- and low-coverage regions look much more comparable.
- Figure 2 shows the key descriptive fact: before reform, low-coverage regions had substantially higher fertility; after reform, the gap closes.
- The age-specific panel of Figure 2 shows that the convergence is concentrated at ages 30 to 45.
- The cohort panel shows that completed fertility differs by about one child for the older cohorts, but the gap shrinks and disappears for younger cohorts exposed to the new pension system.

Economic message. The descriptive figures already point toward the paper’s mechanism: pensions mattered not because women started having children later, but because they stopped earlier.

5.3 Table 4 and Table 5: identification works only after the right exclusions, and the baseline effect is sizable

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TABLE 4—FERTILITY TRENDS BEFORE, DURING, AND AFTER THE EXTENSION

Sample	Include Whites and Khomas (1)	Exclude Whites, include Khomas (2)	Exclude Whites and Khomas (3)
Before extension (1990–1992)	–0.285 (0.435)	–0.200 (0.473)	–0.556 (0.438)
During extension (1992–1997)	–0.769 (0.104)	–0.861 (0.114)	–0.211 (0.311)
After extension (1997–2012)	–0.065 (0.046)	–0.017 (0.034)	–0.018 (0.123)
Before × Whites	–0.003 (0.001)		
During × Whites	0.311 (0.163)		
After × Whites	0.025 (0.073)		
Before × Khomas		–0.002 (0.001)	
During × Khomas		0.405 (0.352)	
After × Khomas		–0.174 (0.048)	
Before × InitialNeeds			0.004 (0.004)
During × InitialNeeds			–1.646 (0.766)
After × InitialNeeds			–0.006 (0.350)
Observations	180,473	155,774	141,637
Clusters	51	38	35

Notes: We estimate the linear annual trend in birth rate separately for three periods: 1990–1992, 1992–1997, and 1997–2012. We multiply coefficients by 100 to improve the readability of the table. 1992 to 1997 is the period of transition from the old pension system, with unequal benefits and coverage, to the new system, with equalized replacement rate and universal coverage. *Whites* is a dummy for people defined as White, Coloured, or Baster during

TABLE 5—PERIOD ANALYSIS: BIRTH RATE

	Dependent variable: probability of birth						
	Baseline (1)	+ urban (2)	+ wealth (3)	+ access to services (4)	+ cost of children (5)	+ AIDS and mortality (6)	+ indiv. controls (7)
Post*InitialNeeds	–0.123 (0.051)	–0.151 (0.045)	–0.171 (0.051)	–0.170 (0.048)	–0.160 (0.045)	–0.177 (0.045)	–0.173 (0.043)
Post*LowInitialNeeds							–0.008 (0.016)
Post*HighInitialNeeds							–0.031 (0.010)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	141,637	141,637	141,637	141,637	141,637	141,637	141,637
Clusters	35	35	35	35	35	35	35
R ²	0.007	0.007	0.007	0.007	0.007	0.007	0.015

Notes: Level of observation: woman*year. Sample: women aged 15 to 40 in year *t*; White ethnic group and Khomas region are excluded. OLS regression. Standard errors are clustered at the region*ethnicity level. *InitialNeeds* is the difference between the poverty line and the expected annual pension for a given ethnic group in a given region before the extension, in thousand rands (average in the bottom half = 0.28; average in the top half = 0.43). *Post* is a dummy for years after 1992 (date of the Pension Act). Full set of controls include urban (share living in urban areas, share active in the agricultural sector), wealth (poor, savings account), access to services (distance to hospital), cost of children (schooling rate 5–14, share of 10–14 economically active), AIDS, under-5 mortality, indiv. time-invariant controls (maternal education). In column 8 the categories “Low,” “Medium,” and “High” Initial Needs consist of groups with average needs of, respectively, 0.23 thousand rands (corresponding to a pension coverage between 50 and 65 percent), 0.32 thousand rands (coverage between 40 and 50 percent), and 0.43 thousand rands (coverage between 30 and 40 percent). Medium initial needs is the omitted category and the regression includes the full set of controls.

Sources: DHS and NHIES

- Table 4 shows why Whites and Khomas are dropped. They have different pre-trends and post-trends. In the preferred sample, the interaction of pre-trend with initial needs is near zero, while the differential decline shows up only during the reform window.
- Table 5 reports the main period DD estimates. The coefficient on *Post* × *InitialNeeds* is –0.123 without controls and stabilizes near –0.17 once controls are added.
- The fact that the coefficient becomes more negative after adding wealth and urbanization controls indicates that omitted local development trends were masking part of the pension effect.
- Column 8 of Table 5 shows the nonlinearity: the high-needs group falls sharply relative to the medium-needs group, while the low-needs group is statistically indistinguishable from medium needs.

Economic message. Table 4 is the credibility table; Table 5 is the main reduced-form result. Together they say that fertility fell more exactly where the old system had left people the most exposed in old age.

5.4 Figure 5, Table 6, and Figure 6: the timing lines up with the reform and with cohort exposure

Read:

- Figure 5 shows that before reform the coefficient on initial needs is flat and near zero. It becomes increasingly negative during the extension and then stabilizes after 1997–1998.
- Table 6 shows that completed fertility falls more for cohorts with greater exposure, and the interaction becomes stronger once controls are added.
- In the partial cohort analysis, the controlled coefficient on *Exposure* × *InitialNeeds* is –4.58, implying about 0.5 to 0.75 fewer children in high-needs relative to low-needs groups.
- Figure 6 shows the cohort-by-cohort pattern: no effect for cohorts born before 1955, progressively

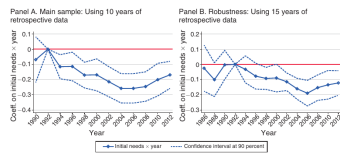


FIGURE 5. PERIOD ANALYSIS—IMPACT OF INITIAL NEEDS ON PROBABILITY OF BIRTH, BY YEAR

Notes: We replace the term $\text{Post}_i \times \text{InitialNeeds}_i$ by $\sum_{j=0}^{10} \text{Post}_{i,j} \times \mathbf{1}\{2 - 1 \leq \text{Year}_i \leq 2\} \times \text{InitialNeeds}_i$ in equation (1), controlling for year and cluster fixed effects and the full set of controls (urban, wealth, access to services, cost of children, AIDS, US monthly, maternal education). We pool together years i and $i-1$ to increase precision. The omitted category is 1991–1992, the last period in which fertility decisions were made before the Pension Act. The graph plots γ_0 and the confidence intervals at 90 percent. Panel A uses the main sample, women aged 15 to 40 in year i , and 10 years of retrospective data covering the period 1990 (start = 1995 to 2012 (end = 1096).

TABLE 6.—COHORT ANALYSIS: COMPLETED FERTILITY

	Dep var: number of births		
	Baseline	Full set of controls	Non-linear effect (3)
	(1)	(2)	(3)
<i>Panel A: Pre-post analysis (1950–1955, 1965–1969)</i>			
Exposure*InitialNeeds	-1.67 (1.76)	-4.98 (1.78)	
Exposure*LowInitialNeeds		0.40 (0.72)	
Exposure*HighInitialNeeds		-0.41 (0.57)	
Cohort FE	Yes	Yes	Yes
Cluster FE	Yes	Yes	Yes
Observations	901	901	901
Clusters	26	26	26
R ²	0.214	0.294	0.293
<i>Panel B: Partial analysis (1950–1969)</i>			
Exposure*InitialNeeds	-1.09 (1.75)	-4.58 (1.63)	
Exposure*LowInitialNeeds		-0.50 (0.72)	
Exposure*HighInitialNeeds		-1.07 (0.51)	
Cohort fixed effects	Yes	Yes	Yes
Cluster fixed effects	Yes	Yes	Yes
Observations	1,798	1,798	1,798
Clusters	28	28	28
R ²	0.132	0.204	0.204

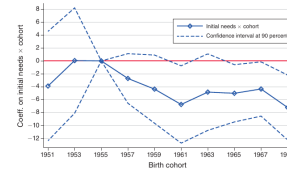


FIGURE 6. COHORT ANALYSIS—IMPACT OF INITIAL NEEDS ON COMPLETED FERTILITY, BY COHORT

Notes: We replace $\text{Exposure}_i \times \text{InitialNeeds}_i$ by $\sum_{j=0}^{10} \text{Exposure}_{i,j} \times \mathbf{1}\{2 \leq \text{Cohort}_i \leq 2\} \times \text{InitialNeeds}_i$ in equation (2), controlling for cohort and cluster fixed effects and the full set of controls (urban, wealth, access to services, cost of children, AIDS, US monthly, maternal education). We pool together cohorts i and $i-1$ to increase precision. The omitted category is 1954–1955, the last cohort not exposed to the extended pension system. The graph plots γ_0 and the confidence intervals at 90 percent.

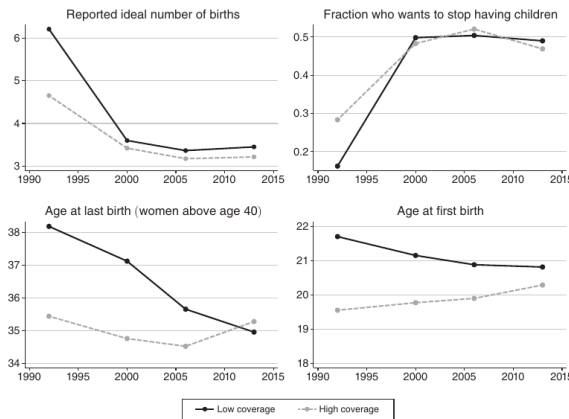


FIGURE 7. MECHANISM—CHANGE IN STOPPING BEHAVIORS

Notes: The graphs show the evolution of the ideal number children (top left), proportion of women who want to stop having children (top right), age at last birth (bottom left) and age at first birth (bottom right) in regions with

more negative effects for cohorts born between 1956 and 1964, and a persistently negative gap for cohorts born after 1965.

Economic message. These dynamics are exactly what one wants to see. The effect does not simply appear sometime after reform; it tracks exposure in a way that is hard to reconcile with a generic secular convergence story.

5.5 Figure 7: the mechanism is stopping, not delayed starts

Read:

- The ideal number of children falls sharply in low-coverage regions and converges to the level in high-coverage regions.
- The fraction of women who want to stop having children rises sharply and converges upward.
- Age at last birth moves down substantially in low-coverage regions and nearly converges fully.
- Age at first birth does not move in the same direction; if anything, the initial gap runs the other way.

Economic message. This is the cleanest mechanism figure in the paper. It shows that pensions reduce desired completed fertility by changing the stopping margin, which is precisely where the

old-age-security motive should bite.

5.6 Nonlinearity and discussion: the effect is strongest when formal protection is very weak

Read:

- The nonlinear specifications in Table 5 and Table 6 show that fertility responds strongly when protection moves from very low to moderate levels, but not much beyond that.
- The discussion section interprets this through a Lexicographic Safety First model: couples first choose enough children to avoid destitution in old age and only then think about standard short-run child costs and benefits.
- In that framework, introducing pensions lowers the minimum target number of children and reduces how sensitive fertility is to the risk that children fail to support parents.

Economic message. The paper's broader conceptual claim is not only that pensions matter, but that they matter most where people are near a subsistence-style safety threshold. That is why the nonlinear effect is so important.

6 Short summary

- The paper studies whether social pensions reduce fertility by weakening the need to rely on children for old-age support.
- It exploits Namibia's post-apartheid equalization of pension benefits and coverage, combining pre-reform variation in pension protection with time and cohort exposure.
- The preferred sample excludes Whites and the Khomas region because those groups violate the parallel-trends logic of the design.
- In both period and cohort specifications, fertility declines more in groups that initially had higher unmet old-age-support needs.
- The effect is concentrated in late reproductive life and operates mainly through stopping earlier, not starting later.
- Mechanism evidence from ideal family size, desire to stop childbearing, and age at last birth points toward a change in desired fertility.
- The fertility response is strongly nonlinear: moving from very low pension protection to moderate protection reduces fertility a lot, while moving from moderate to high protection has little additional effect.
- The paper's external-validity discussion suggests that similar pension expansions elsewhere in sub-Saharan Africa could reduce completed fertility by about one child, and possibly more where there is currently no pension system at all.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that social pensions can significantly reduce fertility. Using the Namibian reform, it shows that women in groups with greater initial needs for informal old-age support reduce fertility more when formal pension protection expands.

7.2 Economic intuition

The central intuition is simple: when old-age consumption depends heavily on children, having more children is a form of insurance. A credible social pension weakens that motive. Because the relevant decision is often whether to keep having children late in life, the response appears mainly on the stopping margin.

7.3 Takeaway

This paper gives one of the clearest quasi-experimental demonstrations that social protection for the elderly can also be a population policy. Pensions do not just redistribute income across generations; they change the incentives that shape fertility itself.